



URBAN FARMING - SACK FARMING TOOLKIT

Table of Contents

1. Introduction to Urban Farming
2. Benefits of Urban Farming
3. Common Methods of Urban Farming
4. Sack Farming: A Practical Guide
 - What is Sack Farming?
 - Why Start Sack Farming?
 - How to Start Sack Farming
 - Tips and Considerations
 - Common Challenges and Solutions
 - What to Plant in Your Sack Farm
5. Conclusion
6. Additional Resources

Introduction to Urban Farming

Urban farming, also known as urban agriculture, refers to the practice of cultivating, processing, and distributing food and other agricultural products within or around urban areas. This innovative approach transforms underutilized spaces in cities such as rooftops, balconies, backyards, and vacant lots into productive agricultural zones. In developing countries like Nigeria, where rapid urbanization and population growth strain traditional food supply chains, urban farming has emerged as a vital strategy for enhancing food security, reducing poverty, and promoting sustainable development.

In Nigeria, urban farming addresses key challenges such as food insecurity, unemployment, and environmental degradation. With over half of the population living in cities, it provides fresh produce to urban dwellers, creates income opportunities for low-income households, and contributes to

dietary diversity. It is particularly important in regions like southeast Nigeria, where surveys of urban farmers highlight its potential to meet growing food demands amid constraints like limited land availability.



Urban farming not only boosts local economies but also fosters community resilience by reducing dependence on imported or rural-sourced foods, especially in volatile economic conditions like those in oil-dependent regions such as Benin City.



Benefits of Urban Farming

Urban farming offers a wide array of advantages, particularly in developing contexts like Nigeria. Here are some key benefits:

1. Food Security and Nutrition: It provides access to fresh, nutritious produce, helping to diversify diets and combat malnutrition among urban populations.

2. Economic Opportunities: Urban farmers can generate income through sales of produce, reducing poverty and creating jobs, especially for women and youth.

3. Environmental Sustainability: By recycling waste, improving air quality, and enhancing biodiversity, urban farming contributes to greener cities and reduces the urban heat island effect.

4. Resource Efficiency: It minimizes transportation needs, conserves water, and promotes soil health, making it a low-impact alternative to traditional agriculture.

5. Community and Health Benefits: Engaging in urban farming promotes physical activity, mental well-being, and community bonding, while also educating people on sustainable practices.

Common Methods of Urban Farming

Urban farming employs various techniques adapted to limited spaces and resources. Common methods include:

1. Rooftop and Vertical Gardening

Utilizing building tops or walls for growing crops in stacked layers.



2. Container Gardening:

Planting in pots, buckets, or recycled containers for easy mobility.



3. Hydroponics and Aquaponics:

Soil-less systems using nutrient-rich water, ideal for indoor or controlled environments.



4. Community Gardens:

Shared plots in public spaces fostering collective farming.



5. Sack Farming:

A space-efficient method using bags or sacks, perfect for balconies or small yards—detailed in the next section.



These methods make urban farming accessible to anyone, regardless of land ownership.

Sack Farming: A Practical Guide

Sack farming is one of the most innovative and accessible methods within urban agriculture, especially in Nigeria where land scarcity is a major issue. This technique allows individuals to grow food in even the smallest spaces, contributing to food security and self-sufficiency.

What is Sack Farming?

Sack farming involves growing plants in sacks or bags filled with soil or other growing media. This method has gained popularity in Nigeria among urban farmers and gardening enthusiasts due to its simplicity and versatility. A wide variety of crops, from vegetables to tubers, can be cultivated using this approach.

Why Start Sack Farming?

Sack farming offers numerous advantages, making it an excellent choice for urban dwellers:

- Saves money on food purchases by providing home-grown produce.
- Ensures easy access to fresh, nutritious food.
- Allows for dietary diversification with a range of crops.

- Optimizes resources, yielding high outputs with minimal inputs like space, water, and time.

Specific Benefits of Sack Farming

Benefit	Description
Space-Saving	Ideal for small areas like balconies, backyards, or urban apartments.
Soil Conservation	No tilling required, reducing erosion and degradation.
Water Efficiency	Uses less water than traditional methods, suitable for water-scarce regions.
Increased Yield	Better soil aeration leads to higher crop production.
Year-Round Production	Enables continuous growing regardless of seasons.

How to Start Sack Farming

Setting up a sack farm is straightforward and requires minimal materials. Here's what you'll need:

Sacks or Bags: Clean, food-safe options like polyethylene, jute, or fiber bags (e.g., recycled rice bags).

Growing Medium: Rich soil or a planting mix.

Fertilizer/Manure/Compost: For nutrient enrichment.

Seeds, Seedlings, or Cuttings: Depending on your chosen crops.

Step-by-Step Guide

1. Clean your sacks thoroughly to remove any contaminants.
2. Place a layer of pebbles or stones at the bottom for drainage.
3. Mix soil and manure/compost in a 1:1 ratio. Add this mixture in layers, watering each layer to settle it.
4. Puncture holes on the sides and bottom of the sacks for aeration and drainage.
5. Plant seeds, seedlings, or cuttings on the top. Optionally, create planting holes on the sides for additional plants.
6. Set up a simple drip irrigation system or water manually with cans as needed.
7. Follow crop-specific practices, such as applying fertilizers, pesticides, or herbicides; pruning; and staking.



Tips and Considerations

To maximize your sack farm's success:

- **Monitor Temperature:** Sacks can heat up; provide shading and ventilation as needed.
- **Pest Management:** Inspect regularly for pests and use natural remedies.
- **Fertilization:** Opt for organic fertilizers to support healthy growth.
- **Plant Support:** Use stakes for climbing plants like tomatoes or peas.
- **Record-Keeping:** Track planting dates, yields, and issues for continuous improvement.

Common Challenges and Solutions

Challenge	Solution
Waterlogging	Add drainage holes or incorporate perlite/vermiculite into the soil mix.
Pests	Apply neem oil, introduce beneficial insects, or use appropriate pesticides.
Diseases	Remove infected plants, improve air circulation, and apply organic treatments.

What to Plant in Your Sack Farm

Sack farming excels with quick-maturing crops that thrive in confined spaces. Recommended options for Nigeria include:

- **Vegetables:** Cucumbers, peppers, tomatoes, leafy greens like Ugu (fluted pumpkin).
- **Tubers:** Yams or potatoes.

These crops have proven to yield bountiful harvests in urban settings.

Conclusion

Urban farming, with techniques like sack farming, represents a transformative solution for food security and sustainability in Nigeria. By adopting these practices, individuals and communities can achieve self-reliance, improve nutrition, and contribute to a greener future.

Start small, grab a sack, fill it with soil, and begin your journey today.

Written By:
Adegbuyi Adewale Michael (M. Agric)
Ikore International Development Limited